

Understanding Smartphone Notifications' Activity Disruption via In Situ Wrist Motion Monitoring

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ABSTRACT

The disruptive nature of smartphone notifications and their negative impact on users' productivity are well documented. The majority of these results either originate from controlled laboratory studies, or protocols relying on subjective self-reporting, reducing their ecological validity. This paper presents results from a full day *in situ* study investigating the impact of perceiving one's smartphone notifications on wrist motion patterns. Through this objective behavioral assessment, we document for the first time the manifestations of notification-induced disruption outside of the lab, independently of user activity and without the need for self-reporting. We identified a decrease in wrist motion activity following the presentation of a notification while the participant was engaged in higher intensity activities, independently of whether the notification is immediately attended to. These findings provide objective support for the claim that notifications have as much potential for disruption when merely perceived as they do when the user actually responds to them.

CCS CONCEPTS

• Human-centered computing → Smartphones.

KEYWORDS

Notifications, perception, smartphone addiction, *in situ* study, wrist motion

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1 INTRODUCTION

While there is little doubt that notifications facilitate effective human-computer and computer-mediated human-human interactions, their current pervasiveness is also known to have significant negative effects on smartphone users' workplace performance and mental well-being. Indeed, frequent interruptions caused by notifications have been shown to negatively impact productivity, attention and the ability to focus [5]. In addition, the unpredictable outcome and presentation times of notifications have been theorized to contribute to the reinforcement of problematic smartphone usage patterns, frequently associated with significant disruption of users' ability to engage in their usual daily activities and social interactions [6, 16].

The vast majority of the literature uses two distinct approaches to assess the interruption burden of smartphone notifications. The first relies on self-reporting, while the second attempts to capture the effect of the interruption via controlled task performance. Although not exhaustive, the following paragraphs presents several examples of these approaches to highlight their advantages and limitations.

Multiple self-reporting methodologies have been used to assess the interruption burden of notifications, often focusing on different cognitive constructs. Okoshi et al. conducted a laboratory study, where participants were presented with notifications originating from different applications while they were engaged in an email writing task [7]. In their study, different notification strategies were employed. Their impact on the user's cognitive load was evaluated by administering a NASA Task Load Index (NASA-TLX) questionnaire after each trial. Kushlev et al. conducted an *in situ* study during which participants were asked to turn off all possible notifications for a full week. Using a daily and weekly questionnaire, they showed that withdrawing from notifications resulted in lower perceived notification-caused distraction, inattention, symptoms of hyperactivity, and higher productivity and general psychological well-being [5]. Weber et al. proposed Annotif, a tool that allows the annotation of notifications at a later time after their presentation [18]. To minimize the effect of retrospective bias, Annotif presents contextual information on when the alert was initially presented (e.g., location, activity). Using their instrument, researchers are able to retrospectively prompt participants about their subjective experience of an alert. The self-reporting approach has the advantage that it directly measures a specific construct of interest (e.g., cognitive load, attention, notification importance). However, it risks introducing biases in the data (e.g., social desirability, acquiescent responding, retrospective) that may negatively impact the quality of the results. In addition, presenting notifications to draw participants towards self-reporting instruments may directly

increase their overall notification volume, influencing the very phenomenon a study initially aims to measure.

The second approach to the study of notifications' disruptiveness relies on objective performance metrics. For example, Iqbal and Horvitz studied the impact of participants receiving work email notifications on the time spent engaged in their primary activities. While they found that email push notifications generated shorter interruptions than having to manually verify whether they had received new emails, the fact that the results are tightly tied to the work email scenario reduces their generalizability to daily personal notifications [4]. Instead of an ecologically valid activity, Stothart et al. asked participants to engage with the Sustained Attention to Response Task (SART) in a laboratory setting [14]. Participants were receiving both messages and calls directly to their own smartphone, and were initially unaware that the notifications were associated with the experiment in which they were participating. The authors reported that receiving a notification can have as much of a detrimental impact on the SART performance as directly attending to the notification itself, either by answering a text message or having a phone discussion. Relying on objective behavioral metrics reduces the risks of introducing biases associated with subjective data collection. However, researchers often have to constrain the context in which they are deployed to ensure the interpretability of the results such as in the examples above (i.e., work emails, laboratory settings).

Supporting this prior literature, this paper presents the first *in situ* evidence that perceiving a notification disrupts activity execution to an extent that significantly alters the user's wrist motion patterns while conducting their regular activities, *even when the user does not immediately attend to the notification*. The observed effect on wrist motion supports the claim that notifications have as much potential for disruption when merely perceived as they do when the user actually responds to them. We also report on how these changes in wrist motion intensity are affected by the user's activity level and notification responsiveness.

2 EXPERIMENT

The central objective of this work is to investigate whether notification-induced disruptions manifest themselves through users' motor behavior, and how these manifestations vary based on users' activity level and responsiveness.

2.1 Protocol

2.1.1 Preparation. Due to constraints imposed by the Covid-19 pandemic, an equipment package containing participant instructions and a Shimmer3 GSR+ sensor was delivered to participants on the day before data collection. A video call was held to walk them through the process of putting on the sensor, as well as downloading, installing and using the data collection application. During that first meeting, they were also asked to read and digitally sign an institutionally approved consent form (REB# 83-0814), and to complete a pre-experiment questionnaire aimed at collecting basic demographic information.

2.1.2 Data Collection. Participants were instructed to use the data collection system for a duration of 8 waking hours, spent in their residence, during which they would be engaging in their regular

activities. Furthermore, participants were told that they had to have their smartphone with them at all time to prevent connection interruptions with the sensor. During data collection, their device's ringer mode had to be set to a non-silent mode, i.e., sound, vibration or sound and vibration, to ensure they would perceive their incoming notifications. As they had their smartphone with them at all time, and its ringer was non-silent, all presented notifications were assumed to be perceived by participants. Beyond these requirements, participants were instructed to use their smartphone as they would on any regular day. At the end of the data collection period, participants used the application to compress and upload their data to a cloud storage service.

2.2 Data Collection and Measurements

2.2.1 Mobile Application. Inspired by prior notification research [2, 8, 9, 17, 18], an Android application was designed to collect information on notification presentation, interactions and to log sensor data. The application saved the data to the device's internal memory to avoid excessive data plan usage. Since Android's notification channel is used by a number of processes that do *not* produce alerts, notification events were filtered [2, 9]. Only notifications that originated from an application that produces alerts, as defined by Iqbal and Bailey [3], and that were presented more than 500 ms after the previous one by a same application were considered for analysis. For each notification, its presentation time, removal time, total active time, reason for removal, source application, screen-lock state at the time of presentation and removal were logged. To protect participants' privacy the content of the notifications was not recorded.

Participants were instructed to wear the sensor¹ on their non-dominant wrist as they would typically wear a watch. Raw sensor data were collected, streamed to the participant's own Android device and logged to its internal memory at a fixed rate of 64 Hz. Accelerometer measurements were first low-pass filtered (zero lag filtering, 8th order Butterworth, 10Hz) to only retain information pertinent to the participants' wrist motion [11]. The three measured axes were then aggregated by computing the norm of the acceleration vector. The mean and spectral centroid of the signal were extracted within a 15 s window before and after each notification presentation. The length of the window was arbitrarily chosen as being representative of short-term changes in activity execution. This simple set of descriptive features should be sufficient to capture changes in wrist motion amplitude and pattern, while remaining easily interpretable. To ensure comparability across notifications between participants and highlight relative changes caused by the presentation of notifications, the percentage change between all pre- and post-notification feature values was computed and used for subsequent reporting and analysis.

2.2.2 Activity Level and Responsivity. Knowing with certainty the activities in which users were engaged in can be particularly challenging when conducting in the wild research. As such, instead of focusing on discrete activities, we instead turned to the users' overall activity level [1] with the assumption that the impact of a notification on wrist motion patterns should be relatively consistent

¹Shimmer3 GSR+, Shimmer Engineering

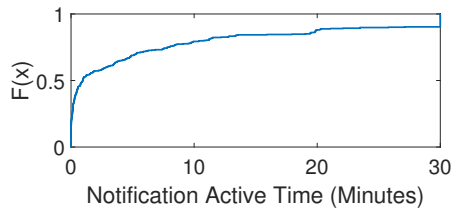


Figure 1: Cumulative frequency distribution of the time for which notifications had been actively displayed.

within low and high relative intensity activities. We hypothesized that perceiving a notification while engaged in a relatively high intensity physical activity (e.g., walking) would result in a decrease in overall wrist motion (i.e., decrease in mean and spectral centroid) as a consequence of the reallocation of attention to the incoming stimulus and the known negative influence of task-irrelevant thoughts on primary task performance [13]. Alternatively, a notification presented in a low activity scenario (e.g., watching a video) might instead result in a relative increase in wrist motion associated with the action of reaching out to and attending to the notification (i.e., increase in mean and spectral centroid). Each notification was assigned to the low or high activity group depending of whether its mean pre-notification acceleration magnitude was under (low) or above (high) the median of each participant’s mean pre-notification acceleration magnitudes.

Since we suspected that the action of attending to a notification (e.g., taking out and manipulating their phone) could mask more subtle subconscious changes in motion behavior, we analyzed notifications that were removed by a user *during* the 15 s post-notification measurement window (immediate) separately from those that were attended to at a later time (delayed). From here on by, this distinction will be referred to as *responsivity*.

2.3 Participants and Compensation

Participants were recruited from university classified ads and social media groups. They were given a base compensation of \$ 5 for their time. A bonus \$ 30 was offered if they complied with the full data collection protocol.

3 RESULTS AND DISCUSSION

3.1 Participants and Notifications

Despite initially recruiting 9 participants, the data from 3 had to be rejected due to unforeseen incompatibility between their smartphone, the data collection application and the Shimmer3 GSR+ sensor. Of the remaining 6 participants, 4 identified as female and 2 as male. Participants reported being between 21 and 28 years old ($\bar{x} = 24.6$, $\sigma = 2.5$). Their primary occupations were: researcher at a local university (1), undergraduate or graduate student (4) and retail worker (1).

Participants received a median of 47 notifications during the 8 hours of data collection (min=9, max=80), in line with the notification volumes reported in prior work, which ranged from 45 to 63 daily notifications [8, 9, 12]. A total of 257 notifications were considered for analysis.

Based on the difference between presentation and removal times, notifications were attended to at a rate similar to that reported in the literature [8, 9, 12]. Indeed, 50% of presented notifications were active for slightly less than 1 minute, 75% less than 7.8 minutes and a total of 90.2% of notifications spent less than 30 minutes in the notification tray (Figure 1) [8]. The remaining 9.8% of notifications were cleared after 30 minutes, or remained active until the end of the study. Unsurprisingly, a Mann-Whitney U test revealed that notifications presented while the screen was unlocked (60.5% of notifications) were removed significantly faster than those presented from a locked screen ($z=-2.86$, ranksum=35912.5, $p=.004$), with median active times of 31.4 seconds and 61.7 seconds respectively.

Participants reported most frequently using their smartphone in silent mode (4/6), followed by sound (1/6) or vibration only (1/6) during their daily life. The least frequently used ringer mode was vibration *and* sound. As such, participants who most often use their device in silent mode had to modify their usual usage pattern for the study, negatively impacting the external validity of the findings presented in this work.

3.2 Impact of Notification Perception on Wrist Motion

Day to day experience and prior literature has shown that the presentation of a notification significantly disrupts users while they are engaged in a primary activity [14]. Researchers typically demonstrate such effects by closely monitoring participants’ behavior while they are engaged in controlled tasks [14], or by using self reporting to determine the degree to which notifications were interrupting [7]. In this work, we were instead interested in objectively measuring how a notification’s interruption manifests itself through changes in users’ wrist motion in the wild, and how users’ activity level and responsivity influences these changes.

Despite our intuition that the perception of a notification would reflect itself differently depending on the activity a user is engaged in, we first conducted an exploration of the impact of notification perception of wrist motion patterns, without differentiating for user activity level and responsivity. To do so, we employed a series of one-sample Wilcoxon signed rank tests to compare the distribution of each wrist motion feature’s median against a percentage change of 0. A median that is statistically significantly different from 0 would be indicative of a change in the manner a user is carrying out an activity after the presentation of a notification. A negative change would suggest a decrease in overall wrist motion, whereas a positive change would instead hint that the perception of a notification caused an increase in the user’s wrist motion intensity. No statistically significant difference between the median of the percentage change of each feature and 0 was observed at the 95% confidence level. As per our previously stated hypothesis, it is possible that the changes in wrist motion activity following the presentation of a notification vary in direction when considering high and low intensity activity, cancelling out each other over extended periods of time. For example, a jogger might slow down their pace momentarily, voluntarily or not, to reallocate their attention towards a notification. Later, that same person might be relaxing on their sofa and reach out to their smartphone after the

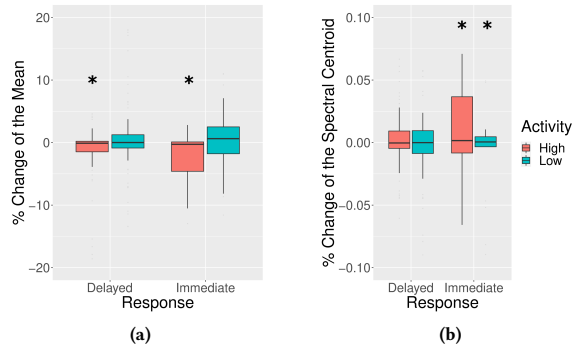


Figure 2: (a) Percentage change of the mean acceleration magnitude of the wrist, showing a statistically significant decrease in motion in the high activity condition. (b) Percentage change of the spectral centroid. Asterisks are used to identify medians that statistically significantly differ from 0.

	F	Df	Df.res	P-value	η_p^2
Activity	11.21907	1	251.13	<.001	0.04
Responsivity	1.38712	1	195.65	0.24	0.01
Activity:Responsivity)	0.13313	1	249.84	0.72	0.00

Table 1: Results from the ARTANOVA investigating the effect of user activity level and responsivity on the percentage change in mean wrist acceleration computed between before and after the presentation of a notification.

	F	Df	Df.res	P-value	η_p^2
Activity	2.6643	1	249.61	0.10	0.01
Responsivity	15.5396	1	238.64	<0.001	0.06
Activity:Responsivity)	12.5955	1	249.02	<0.001	0.05

Table 2: Results from the ARTANOVA investigating the effect of user activity level and responsivity on the percentage change in position of the spectral centroid computed between before and after the presentation of a notification.

presentation of a notification, resulting in a relative increase in wrist motion.

Repeated measures aligned ranks transform analyses of variance (ARTANOVA) were employed to investigate the effect of users' activity level and responsivity on the two selected wrist-motion features (Table 1-2). Partial Eta squared are reported as measures of effect size. A second series of one-sample Wilcoxon signed rank tests was used to determine whether a statistically significant change in motion pattern occurred for each condition combination (activity and responsivity) following the presentation of a notification (i.e., deviation from 0% change).

As per Table 1, results suggest that the perception of a notification has a statistically significantly different effect on a user's

wrist motion depending on the intensity of the activity they are engaged in. A statistically significant difference in median between the two activity levels was observed for the percentage change of the mean wrist acceleration, but not for the percentage change of the spectral centroid. In addition, the median of the percentage change of the mean wrist acceleration was found to be lower than 0 for the cases where participants were engaged in higher intensity activities ($V_{HighDelayed} = 1388, p_{HighDelayed} = .01, V_{HighImmediate} = 123, p_{HighImmediate} < .01$). Combined, these results suggest that the perception of a notification has a significant impact on wrist motion amplitude, and that in cases where a user is engaged in a more intense activity, the presentation of a notification will typically result in a statistically significant decrease in overall wrist motion. A decrease in mean wrist motion was observed in both the immediate and the delayed response scenarios, suggesting that the perception of a notification, even when not immediately attended to has a significant impact on how activities are being carried out, as assessed by wrist motion activity. Despite the inability to ascertain the user behavior behind these results given the nature of in-the-wild studies, we offer two possible explanations for the reported decrease in activity following the presentation of a notification:

- *General activity disruption:* The perception of a notification momentarily distracts a user, without necessarily detracting them from their primary activity. This may influence their primary task performance [13, 14] to an extent that is measurable via a wrist-worn accelerometer. In addition, the activity disruption could be accompanied by stereotypical gestures associated with phone checking behavior such as touching, looking at and/or unlocking their smartphone, without necessarily going through the full attendance of a notification which would result in its removal from the notification tray [10]. This hypothesis would be in line with the existing literature based on laboratory studies that reported decrease in task performance and completion time when participants were presented with alerts and notifications [14]. This general activity disruption hypothesis could explain the decrease in wrist motion activity for notifications for which attendance was delayed, since the actions and gestures associated with the full removal of the alert would not be accounted for in the 15 s accelerometer measurement window.
- *Full notification consumption:* In addition to the general activity disruption caused by notifications, the actions associated with the full removal of a notification, i.e., stopping or reducing the intensity of an activity to hold the smartphone, clearing the notification from the notification tray and/or opening the application that generated the notification, should significantly alter wrist motion patterns. While there is no statistically significant difference between the immediate and delayed responses, immediate responses visibly induced more variable changes in wrist motion activity than their delayed counterpart (Figure 2)

Of the features considered in this study, only the percentage change of the spectral centroid was found to statistically significantly vary based on whether a user would attend to a notification within a 15 s window after the presentation of a notification (Table 2). That being said, the statistically significant interaction between activity and responsivity, combined with a median percentage change inferior to 0.01% suggest that the observed results, albeit statistically significant do not offer practically meaningful insights. Contrary to our initial hypothesis, our results do not suggest that users' wrist patterns change practically meaningfully depending on whether the user delays, or immediately attend to an incoming notification. Knowing that decreases in mean wrist motion were previously reported for "high" intensity activity in both responsivity conditions, this supports the argument that the disruptive impact of a notification on user activity engagement is *independent of whether users immediately engage with a notification*. As such, results from our *in situ* study based on participants' own smartphone, device usage habits and notifications objectively supports prior findings from Stothart et al. that had identified the same effect in laboratory conditions [14].

4 LIMITATIONS

Data collection for this study was executed amidst a global pandemic. As such, even though the measurements were made *in situ*, the fact that participants spent most of their day at home is not representative of a "normal", as in pre-COVID, daily amount of physical activity. Public health measures and the negative affects caused by the pandemic itself may have also significantly impacted participants' psychological state. Indeed, a recent study by Varma et al. reported that a large proportion of young adults were experiencing severe stress, anxiety, depression and general psychological distress since the beginning of the pandemic [15]. This may have changed participants' response patterns and behaviors. Beyond its impact on the generalizability of the findings, the pandemic and the public health measures in place also significantly impacted the participant recruitment and experiment execution process. While this is beyond the scope of this work, we hypothesize that participant recruitment for this study was hampered relative to similar experiments in previous years, potentially due to an increase in perceived risks of virus propagation. In addition, for most of the data collection, the city where the study took place was under an 8 pm to 5 am curfew, which extended the study participation to 3 days, with only one day used for data collection. This may have negatively influenced the perceived value of the monetary compensation for participants, further impeding the recruitment process. As such, this paper reports on data collected from only 6 individuals. Even though we believe we have collected a sufficient amount of notifications to conduct meaningful preliminary statistical analyses, the results cannot be generalized to the global population. This is particularly true given the limited age range of participants.

Despite being instructed to set their smartphone to a non-silent ringer mode and to keep it close to them at all times, it is possible that some notifications were not perceived by participants. For example, they may have been fully immersed in a primary activity and not feel or hear a notification. Alternatively, listening to music, watching a video or engaging in other sensory activities might

have masked the signals. That being said, we believe that this is a small price to pay in comparison to the loss of ecological validity that one would encounter when studying behavioral responses to notifications in laboratory settings [7, 14].

5 CONCLUSION

This paper presented the first behavioral evidence of notification-induced activity disruption outside of the lab. Through a full day *in situ* study, we have shown that the perception of a notification results in a statistically significant decrease in wrist motion when users are engaged in a "high" intensity activity, regardless of whether they delay, or immediately attend to an incoming notification. These results, acquired using participants' own smartphones and notifications, objectively support the prior literature on the topic and highlight the significant risks associated with the mere presentation of notifications while engaging in safety critical activity such as driving, operating machinery or conducting a surgery.

Beyond providing us with a better understand of the impact of notifications on smartphone users in the wild, it is possible that the observed changes in wrist motion following the presentation of a notification could be employed to determine whether a notification has been perceived, without the need for users to interact with their smartphone. Such an application could extend the perception-aware smart notification framework introduced by Fortin et al. that was based exclusively on notification-induced changes in physiological signals [2].

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REFERENCES

- [1] Jeffrey R. Blum, Ilja Frissen, and Jeremy R. Cooperstock. 2015. Improving Haptic Feedback on Wearable Devices through Accelerometer Measurements. In *Proceedings of the 28th Annual ACM Symposium on User Interface Software & Technology* (Charlotte, NC, USA) (UIST '15). Association for Computing Machinery, New York, NY, USA, 31–36. <https://doi.org/10.1145/2807442.2807474>
- [2] Pascal E. Fortin, Elisabeth Sulmont, and Jeremy Cooperstock. 2019. Detecting Perception of Smartphone Notifications using Skin Conductance Responses. In *Conference on Human Factors in Computing Systems - Proceedings*. ACM, New-Orleans, LA, 9. <https://doi.org/10.1145/3290605.3300420>
- [3] Shamsi T Iqbal and Brian P Bailey. 2005. Investigating the Effectiveness of Mental Workload As a Predictor of Opportune Moments for Interruption. In *CHI '05 Extended Abstracts on Human Factors in Computing Systems (CHI EA '05)*. ACM, New York, NY, USA, 1489–1492. <https://doi.org/10.1145/1056808.1056948>
- [4] Shamsi T Iqbal and Eric Horvitz. 2010. Notifications and awareness: a field study of alert usage and preferences. In *Proceedings of the 2010 ACM conference on Computer supported cooperative work*. 27–30.
- [5] Kostadin Kushlev, Jason Proulx, and Elizabeth W. Dunn. 2016. "Silence your phones": Smartphone notifications increase inattention and hyperactivity symptoms. In *Conference on Human Factors in Computing Systems - Proceedings (CHI '16)*. ACM, New York, NY, USA, 1011–1020. <https://doi.org/10.1145/2858036.2858359>
- [6] Yu-Hsuan Lin, Chih-Lin Chiang, Po-Hsien Lin, Li-Ren Chang, Chih-Hung Ko, Yang-Han Lee, and Sheng-Hsuan Lin. 2016. Proposed Diagnostic Criteria for Smartphone Addiction. *PLoS one* 11 (11 2016), e0163010–e0163010. Issue 11. <https://doi.org/10.1371/journal.pone.0163010>
- [7] T Okoshi, J Ramos, H Nozaki, J Nakazawa, A K Dey, and H Tokuda. 2015. Attelia: Reducing user's cognitive load due to interruptive notifications on smart phones. In *2015 IEEE International Conference on Pervasive Computing and Communications (PerCom)*. 96–104. <https://doi.org/10.1109/PERCOM.2015.7146515>

- [8] Martin Pielot, Karen Church, and Rodrigo De Oliveira. 2014. An in-situ study of mobile phone notifications. In *MobileHCI 2014 - Proceedings of the 16th ACM International Conference on Human-Computer Interaction with Mobile Devices and Services (MobileHCI '14)*. ACM, New York, NY, USA, 233–242. <https://doi.org/10.1145/2628363.2628364>
- [9] Martin Pielot, Amalia Vradi, and Souneil Park. 2018. Dismissed! A detailed exploration of how mobile phone users handle push notifications. In *MobileHCI 2018 - Beyond Mobile: The Next 20 Years - 20th International Conference on Human-Computer Interaction with Mobile Devices and Services, Conference Proceedings (MobileHCI '18)*. ACM, New York, NY, USA, 3:1–3:11. <https://doi.org/10.1145/3229434.3229445>
- [10] Martin Pielot, Amalia Vradi, and Souneil Park. 2018. Dismissed! A Detailed Exploration of How Mobile Phone Users Handle Push Notifications. In *Proceedings of the 20th International Conference on Human-Computer Interaction with Mobile Devices and Services (Barcelona, Spain) (MobileHCI '18)*. Association for Computing Machinery, New York, NY, USA, Article 3, 11 pages. <https://doi.org/10.1145/3229434.3229445>
- [11] Sander Schreven, Peter J. Beek, and Jeroen B.J. Smeets. 2015. Optimising filtering parameters for a 3D motion analysis system. *Journal of Electromyography and Kinesiology* 25, 5 (2015), 808–814. <https://doi.org/10.1016/j.jelekin.2015.06.004>
- [12] Alireza Sahami Shirazi, Niels Henze, Tilman Dinger, Martin Pielot, Dominik Weber, and Albrecht Schmidt. 2014. Large-scale assessment of mobile notifications. In *Conference on Human Factors in Computing Systems - Proceedings (CHI '14)*. Association for Computing Machinery, New York, NY, USA, 3055–3064. <https://doi.org/10.1145/2556288.2557189>
- [13] Jonathan Smallwood and Jonathan W Schooler. 2006. The restless mind. *Psychological bulletin* 132, 6 (2006), 946.
- [14] Cary Stothart, Ainsley Mitchum, and Courtney Yehnert. 2015. The attentional cost of receiving a cell phone notification. , 893–897 pages. <https://doi.org/10.1037/xhp0000100>
- [15] Prerna Varma, Moira Junge, Hailey Meaklim, and Melinda L Jackson. 2021. Younger people are more vulnerable to stress, anxiety and depression during COVID-19 pandemic: A global cross-sectional survey. *Progress in Neuro-Psychopharmacology and Biological Psychiatry* 109 (2021), 110236. <https://doi.org/10.1016/j.pnpbp.2020.110236>
- [16] Samuel P.L. Veissière and Moriah Stendel. 2018. Supernatural monitoring: A social rehearsal account of smartphone addiction. *Frontiers in Psychology* 9 (2018), 141. Issue FEB. <https://doi.org/10.3389/fpsyg.2018.00141>
- [17] Dominik Weber, Alexandra Voit, and Niels Henze. 2019. Clear All: A Large-Scale Observational Study on Mobile Notification Drawers. In *Proceedings of Mensch Und Computer 2019 (MuC'19)*. Association for Computing Machinery, New York, NY, USA, 361–372. <https://doi.org/10.1145/3340764.3340765>
- [18] Dominik Weber, Alexandra Voit, Gisela Kollotzek, and Niels Henze. 2019. Annotif: A system for annotating mobile notifications in user studies. In *ACM International Conference Proceeding Series (MUM '19)*. Association for Computing Machinery, New York, NY, USA. <https://doi.org/10.1145/3365610.3365611>